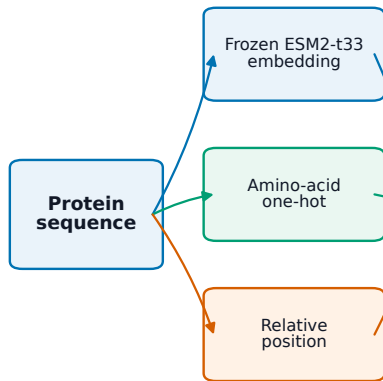
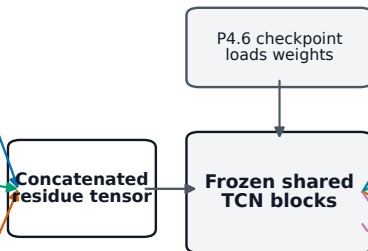


RegionAdapterMoETCN sequence-only prediction workflow

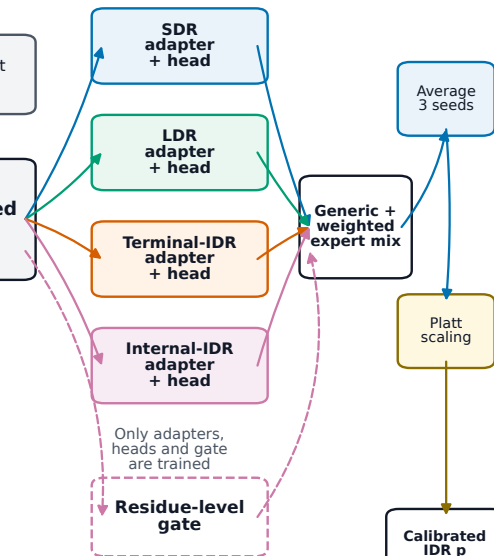
a Residue inputs



b Warm-start backbone



c Region-adapter MoE



d Calibration output

Sequence-only inputs:
no PSSM, MSA, PDB coordinates,
AlphaFold confidence or function labels

Frozen modules preserve the P4.6 backbone;
trainable adapters provide local score correction

Gate weights are reported as auxiliary mechanism evidence,
not as direct biological region annotations